7 SUPPLEMENTAL MATERIAL

A NOTATIONS

The comprehensive list of the primary symbols used throughout this paper is presented in Table 6.

Table 6: Forms and interpretations of notations.

Symbol	Definition
$G = (V, E, X)$	The original graph.
V, E, X	The node set , the edge set, the node feature matrix.
N, M	The number of nodes, the number of edges.
$A \in \{0,1\}^{N \times N}$	The adjacency matrix of the original graph.
f, d	The dimension of feature/embedding.
$\mathcal{Y} \in \{1, \ldots, K\}^n$	Clustering assignments for all nodes.
K	Number of clusters.
$\mathcal{T}; \lambda$	Encoding tree; The root vertice of the encoding tree.
λ	Root node of the encoding tree $\mathcal{T}$.
$\alpha; \alpha^-$	Vertice on encoding tree. Parent vertice of vertice α.
T_α	Subset of graph nodes corresponding to tree node α.
g_α	Total weight of cut edges between T_α and $V \setminus T_\alpha$.
d_v	Degree of node v.
$vol(G); vol(\alpha)$	Volume of Graph G; Volume of vertice α.
$H^{\mathcal{T}}(G)$	Structural entropy of graph G induced by encoding tree $\mathcal{T}$.
h	The height of the encoding tree.
$C = \{c_1, \ldots, c_N\}$	Set of communities produced by SECC.
$c(v)$	Community assignment of node v.
N_{comm}	Target number of communities in SECC.
p	Parallel merge ratio controlling agglomeration rate.
$\Delta H(c_i, c_j)$	Change in SE when merging communities c_i and c_j.
OP	Number of candidate community pairs that could merge.
$\mathcal{V}_{batch}$	Initial set of sampled nodes in a mini-batch.
$\mathcal{V}_{ext}$	Community-guided expansion node set in CSampE.
$\mathcal{V}^*_{batch}$	Final expanded batch after CSampE.
θ	Community size threshold for batch expansion strategy.
w_l, w_t	Length and number of random walks, respectively.
$f(u; v)$	Random walk visitation frequency from v to u.
$\mathcal{R}_v$	Reachable node set of v via intra-community walks.
$\mathbf{A}^* \in \mathbb{R}^{B \times B}$	Structural affinity matrix constructed within a mini-batch.
$\mathbf{z}_i$	Latent embedding of node v_i.
$\mathbf{Z}_{batch}$	embeddings of nodes in a mini-batch.
$\mathbf{Z}_{global}$	consolidated global node embeddings.
$\mathcal{P}_i$	Structural positive set of node v_i.
$\mathcal{L}_{struct}$	Structural contrastive learning objective.

B DATASET

Detailed descriptions of 6 datasets are provided below:

• **Photo** [28] is a segment of the Amazon co-purchase graph specifically focusing on the photography category. Nodes represent products (e.g., cameras and lenses), and edges indicate that two items are frequently bought together. Each node is associated with a 745-dimensional feature vector derived from product reviews in a bag-of-words format, reflecting the fine-grained attributes of photography equipment.

• **Computers** [28] is also a part of the Amazon co-purchase network, where nodes correspond to computer-related products and edges represent frequent co-purchase relationships. Similar to the Photo dataset, the node features are encoded as bag-of-words vectors based on product descriptions and customer reviews, capturing the semantic similarities between electronic components and peripherals.

• **Pubmed** [27] is a citation network consisting of scientific publications related to diabetes from the PubMed database. Nodes represent documents, and edges denote citation links between them. Each node is characterized by a feature vector represented by weighted TF-IDF scores of unique words from the document's abstract.

• **Ogbn-arxiv** [13] is a directed graph representing the citation network between all Computer Science (CS) arXiv papers indexed by MAG. Each node is an arXiv paper, and each directed edge indicates that one paper cites another. The node features are derived by averaging the word embeddings of the paper's title and abstract, and the task is to classify papers into 40 subject areas.

• **Reddit** [12] is a large-scale social network dataset constructed from Reddit posts made in September 2014. Nodes represent posts, and an edge exists between two nodes if the same user commented on both posts. The node features are calculated based on GloVe word vectors, and the communities are defined by the subreddits to which the posts belong.

• **Ogbn-products** [13] is an undirected and unweighted graph representing an Amazon product co-purchasing network. Nodes represent products sold on Amazon, and edges indicate that the products are purchased together. Node features are generated by performing Principal Component Analysis (PCA) on the product descriptions, making it a benchmark for large-scale graph clustering and classification.

C BASELINES

Detailed descriptions of 10 baselines compared to our work are introduced as follows:

• DGI [34] is a self-supervised node representation learning approach that maximizes the mutual information between local patch representations and high-level graph summaries to capture global structural context.

• S3GC [9] is a scalable self-supervised graph clustering framework that employs contrastive learning to integrate node features with topological structures, effectively learning cluster-friendly representations.

• SUBLIME [23] is a structure bootstrapping contrastive Learning framework with the aid of self-supervised contrastive learning, where the learned graph topology is optimized by data itself.

• CONVERT [43] is a contrastive graph clustering network with reliable augmentation and distills reliable semantic information by recovering the perturbed latent embeddings.

• DMoN [33] introduces a modularity measure of clustering quality to optimize cluster assignment in an end-to-end manner and proposes Deep Modularity Networks.

• Dink-Net [20] is an end-to-end scalable deep graph clustering method that optimizes clustering distributions through adversarial dilation and shrink losses to guide the network in learning discriminative features.

• LSEnet [32] is a hyperbolic graph clustering network that formulates a differentiable structural information (DSI) objective, enabling the model to reveal hierarchical cluster structures and determine cluster numbers in a self-supervised manner.

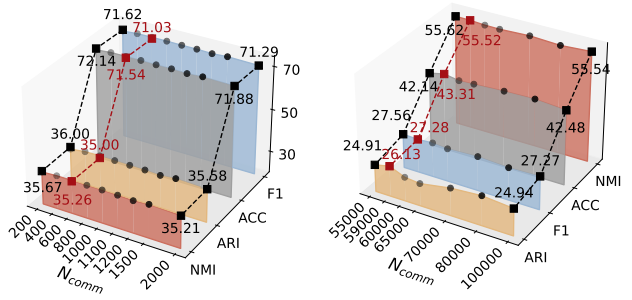
• MAGI [18] is a graph clustering framework that utilizes modularity maximization as a contrastive pretext task to define positive and negative samples, capturing structural information without manual data augmentations.

• DeSE [50] is an unsupervised graph clustering framework that incorporates differentiable structural entropy with soft assignment, utilizing a structural learning layer and a GNN-based assignment module to optimize clustering on an enhanced graph.

• SCGC [15] is a scalable graph clustering framework that eliminates traditional GNN convolutions in favor of simple linear units and dynamic soft structure fusion via an augmentation-less edge-contrastive loss.

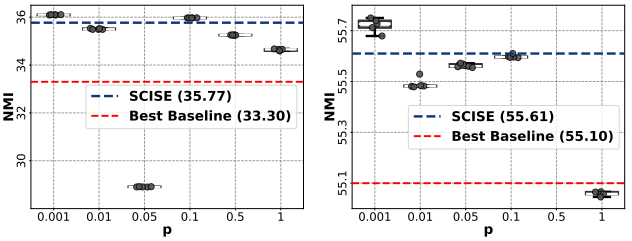
D SENSITIVITY ANALYSIS

This section provides supplementary results for the hyper-parameter sensitivity analysis discussed in Section 5.4. We present additional evaluation metrics (e.g., ACC) and detailed analyses for the random walk parameters (w_t, w_l), further validating the robustness of the SCISE framework.



(a) Pubmed. (b) Ogbn-products.

Figure 11: Analysis of hyper N_{comm} on Pubmed and Ogbn-products datasets.



(a) Pubmed. (b) Ogbn-products.

Figure 12: Analysis of hyper p on Pubmed and Ogbn-products datasets.

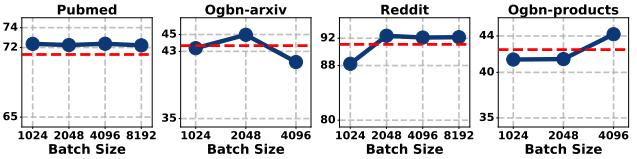


Figure 13: Analysis of hyper B with ACC.

• Impact of community number N_{comm}: Figure 11 illustrates the stability of SCISE on the Pubmed and Ogbn-products datasets as the target community number N_{comm} varies. The flat curves across wide intervals confirm that the SECC operator effectively anchors the partition regardless of the specific N_{comm} value.

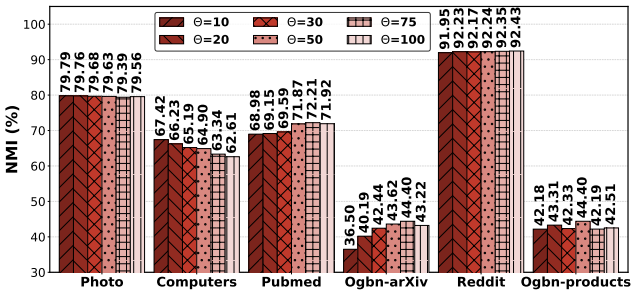


Figure 14: Analysis of hyper θ with ACC.

Table 7: Analysis of hyper w_t with NMI.

w_t	10	20	30	40	50	60	80	100
Photo	73.87	74.00	74.21	74.12	74.40	74.42	74.18	74.15
Computers	56.57	58.40	57.82	56.50	56.54	56.60	56.42	56.50
Pubmed	35.36	35.58	35.62	35.54	35.73	35.21	35.73	35.35
Arxiv	48.28	48.34	48.35	48.50	47.98	48.53	48.35	48.27
Reddit	86.31	87.46	88.35	88.13	88.15	88.08	87.75	87.72
Products	54.63	55.34	55.52	OOM	OOM	OOM	OOM	OOM

Table 8: Analysis of hyper w_l with NMI.

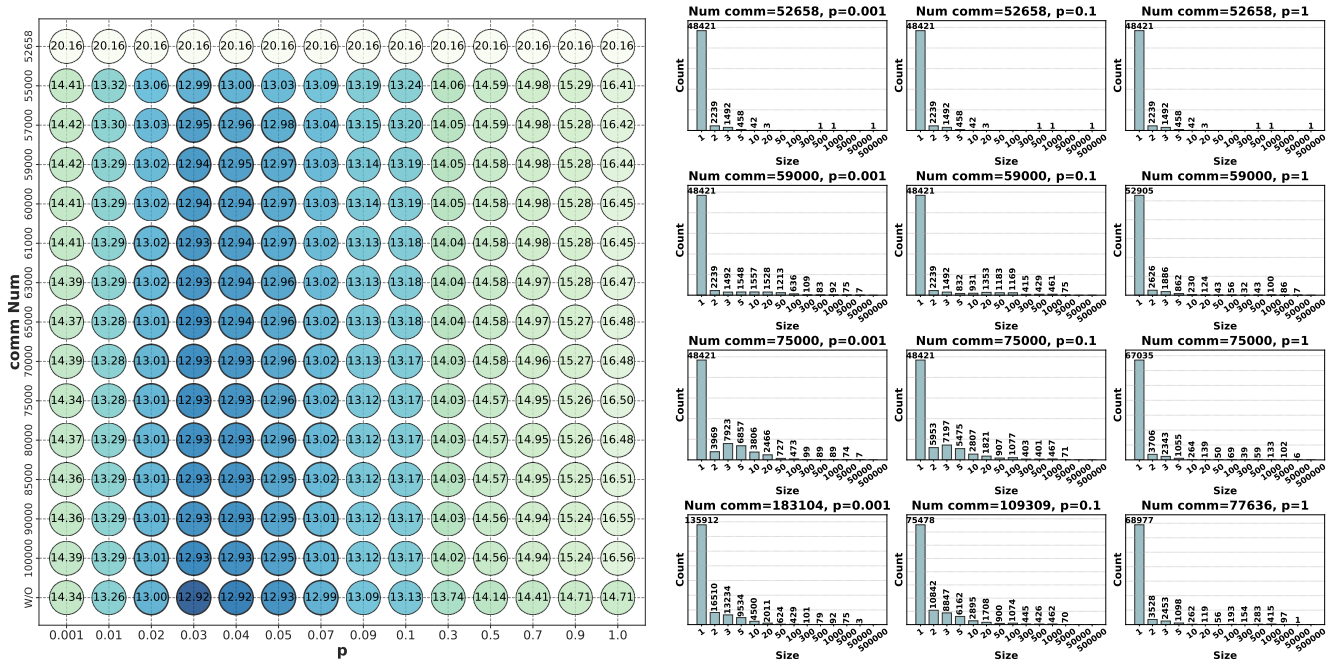
w_l	2	3	4	5	6	7	8	10
Photo	74.49	74.01	73.43	72.57	71.95	71.28	70.88	69.94
Computers	56.20	56.81	57.13	57.26	59.02	58.03	58.52	58.24
Pubmed	35.62	33.67	31.34	30.87	33.25	33.10	33.54	33.99
Ogbn-arxiv	48.60	48.40	48.46	48.01	47.93	47.97	48.02	47.98
Reddit	87.56	88.25	88.28	87.99	87.72	87.34	87.04	86.58
products	54.18	55.24	55.27	55.60	55.77	55.96	OOM	OOM

• Impact of merge speed p: Figure 12 shows the distribution of performance across different merge speeds p on the Pubmed and Ogbn-products datasets. Box plots show that, at different structure entropy minimization granularities, as long as extreme values are not taken, the model performance is always better than the optimal baseline.

• Impact of initial batch size B: Figure 13 shows that the model's accuracy is higher than the baseline for most small batch sizes and remains relatively stable, further demonstrating its scalability and decoupling from hardware limitations.

• Impact of community size threshold θ: In Figure 14, we present the ACC scores across different expansion thresholds θ. Similar to the NMI results, the model remains stable for most benchmarks. Only on the Ogbn-arxiv dataset does it exhibit a slight trend with increasing θ.

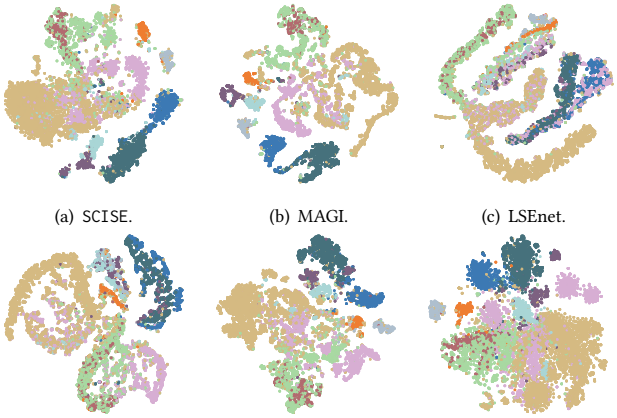
We further investigate the random walk parameters used in the CSampE and StructCL modules, specifically the walk length w_l and the number of walks per node w_t. As shown in Table 7, the performance of SCISE remains remarkably stable as w_t increases from 10 to 100. This suggests that the structural backbone of a community can be effectively captured even with a relatively small number of walks, ensuring computational efficiency. Table 8 details the impact of w_l. While longer walks theoretically provide a broader receptive field, the results indicate that a moderate length is sufficient to capture the essential topological semantic information within cohesive communities.



(a) Heatmap of Structural Entropy on Ogbn-products.

(b) Histogram of community size on Ogbn-products.

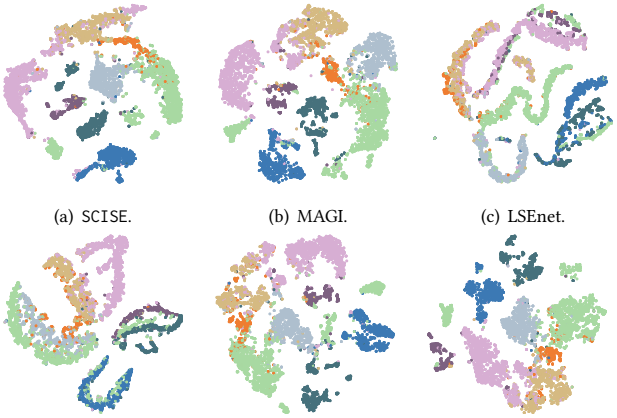
Figure 15: Evolution of structural entropy and community distribution on Ogbn-products.



(a) SCISE. (b) MAGI. (c) LSEnet.

(d) SCGC. (e) DeSE. (f) DinkNet.

Figure 16: Visualization on Computers.



(a) SCISE. (b) MAGI. (c) LSEnet.

(d) SCGC. (e) DeSE. (f) DinkNet.

Figure 17: Visualization on Photo.

Table 9: Hyperparameter settings.

Dataset	N_{comm}	p	θ	batchsize	lr	w_t	w_l
Photo	300	0.1	25	2048	0.005	120	2
Computers	450	0.1	10	1024	0.0005	30	10
Pubmed	600	0.1	70	4096	0.005	30	2
Ogbn-arxiv	3500	0.04	80	2048	0.01	30	3
Reddit	1000	1	70	2048	0.01	30	4
Ogbn-products	59000	0.1	20	4096	0.005	30	5

E HYPERPARAMETER CONFIGURATIONS

For the purpose of reproducibility and experimental transparency, we provide the specific hyperparameter settings for SCISE across six datasets. These configurations are consistent with the experimental results reported in Section 5.2. The detailed parameter assignments are summarized in Table 9.